

1. Stern and Gerlach used the Stern-Gerlach apparatus to test the magnetic orientation of silver atoms. The apparatus had vee-shaped magnets, with the south magnet being stronger. The classical expectation was a continuous line on the screen if magnetic poles could align freely. However, the experiment revealed just two dots on the screen – one at the top and one at the bottom. All silver atoms behaved like vertically aligned bar magnets, defying classical expectations. This indicated quantization of angular momentum, showing that magnetic poles were restricted to specific orientations. The significance lies in challenging classical views and contributing to the development of quantum mechanics, highlighting the discrete nature of atomic and subatomic behavior.
2. True: When the Stern-Gerlach apparatus is rotated, the orientation of the magnetic field changes. Since the interaction with the magnetic field determines the quantized angular momentum projection, changing the orientation of the field will change the observed outcomes of the experiment.
3. Tossing a coin is not a physically random event because in theory if we knew all the initial conditions and forces acting on the coin, we could predict the outcome deterministically. However, in practice, these factors are challenging to measure precisely, and small variations can lead to different results. The outcome is influenced by a combination of factors that are difficult to control or predict precisely. When a coin is tossed, factors such as initial conditions (position, velocity), air resistance, and the interaction with the surface on which it lands contribute to the outcome.
4. A. Since there is a relatively even number of red and blue objects, the percentage would be 50% blue objects and 50% red objects.  
 B. The percentage of objects expected to exit the second filter as red would still be 100% since the filter is coming from output of the red tunnel.  
 C. The percentage of objects expected to be circles would depend on the initial distribution of square and circle objects. Since the initial distribution is relatively even, we can expect 50% of the objects to be circle.  
 D. The percentage of the output expected to be red would be approximately 50%.
5. A. 50% of the light will make it through the first filter because polarizing filters only allow light waves vibrating in a specific direction to pass through. Unpolarized light, like that from an incandescent bulb, vibrates in all directions. The filter blocks half of the light, leaving only the waves aligned with its polarization axis. The intensity of light is 50.  
 B. 100% of the light will still make it through the second filter because the second filter is aligned with the first one, so it doesn't further block any of the light that has already been polarized in the correct direction. The intensity of light is 50.  
 C. The angle between the filter's polarization axes determines how much light passes through. Using Malus's Law, where theta is the angle between the filters, we get  $I = 50 * \cos^2(30) \approx 37.5$ . The intensity will be approximately 38. The percentage difference will be around 27.27%  
 D. The third filter is 15 degrees offset from both the first and second filters. Using Malus's law, we calculate the intensity after the second filter as  $50 * \cos^2(15) \approx 43$ . Then, we apply Malus's law again for the third filter, using 43 as the initial intensity and 15 degrees as the angle:  $I = 43 * \cos^2(15) \approx 30$ .